

Qwen vs Llama Final Correctness Comparison Report

Files: Pipeline_Retest_Report_qwen_all.csv and Pipeline_Retest_Report_llama_all.csv

Report date: 10 June 2026

Purpose: compare Qwen and Llama with reference to the ground-truth answer, check which answers and repeated variants are correct, and verify whether the correctness labels are actually supported by the retest CSV evidence.

Executive Verdict

Qwen is the stronger model in this retest. Qwen has 52 fully correct answers out of 142 rows (36.6%). Llama has 12 fully correct answers out of 143 rows (8.4%). Counting PARTIAL as usable, Qwen reaches 75/142 (52.8%) while Llama reaches 25/143 (17.5%).

Correctness is interpreted from New Status: MATCH = correct against ground truth; PARTIAL = partly correct; MISMATCH and CRASH = not correct. A separate sanity check was also run on JSON, numeric, and explicit no-data ground truths.

Overall Correctness

Qwen: total 142; MATCH 52 (36.6%); PARTIAL 23 (16.2%); MISMATCH 60 (42.3%); CRASH 7 (4.9%); MATCH+PARTIAL 75 (52.8%).

Llama: total 143; MATCH 12 (8.4%); PARTIAL 13 (9.1%); MISMATCH 112 (78.3%); CRASH 6 (4.2%); MATCH+PARTIAL 25 (17.5%).

Head-to-Head on Shared Question Occurrences

The files share 142 question occurrences. Llama has one extra occurrence not present in Qwen. Using the ranking MATCH > PARTIAL > MISMATCH > CRASH:

- Qwen better: 62 (43.7%)
- Tie MISMATCH: 58 (40.8%)
- Tie MATCH: 12 (8.5%)
- Llama better: 4 (2.8%)
- Tie CRASH: 4 (2.8%)
- Tie PARTIAL: 2 (1.4%)

Status-pair counts (Qwen status vs Llama status):

- MISMATCH vs MISMATCH: 58
- MATCH vs MISMATCH: 30
- PARTIAL vs MISMATCH: 21
- MATCH vs MATCH: 12
- MATCH vs PARTIAL: 10
- CRASH vs CRASH: 4
- CRASH vs MISMATCH: 3
- PARTIAL vs PARTIAL: 2
- MISMATCH vs CRASH: 1
- MISMATCH vs PARTIAL: 1

Pipeline Reliability

Qwen also performed better operationally. Llama has 77 scan errors while Qwen has 18. Llama failed most often at Scan or Validate; Qwen still has failures, but many more rows reached successful scan/validation paths.

Qwen: scan success 117, scan error 18, failure Scan 17, failure Validate 51, failure Exception 7.

Llama: scan success 60, scan error 77, failure Scan 55, failure Validate 49, failure

Exception 6.

Old-to-New Retest Movement

Qwen top transitions:

- MISMATCH -> MISMATCH: 57
- MISMATCH -> PARTIAL: 22
- NO_ANSWER -> MATCH: 19
- MISMATCH -> MATCH: 17
- PARTIAL -> MATCH: 16
- MISMATCH -> CRASH: 6
- NO_ANSWER -> MISMATCH: 3
- PARTIAL -> PARTIAL: 1

Llama top transitions:

- MISMATCH -> MISMATCH: 92
- NO_ANSWER -> MISMATCH: 15
- PARTIAL -> MATCH: 7
- PARTIAL -> PARTIAL: 5
- PARTIAL -> MISMATCH: 5
- NO_ANSWER -> PARTIAL: 4
- MISMATCH -> PARTIAL: 4
- NO_ANSWER -> CRASH: 3

Variant / Duplicate Question Check

Both files contain 12 exact duplicate questions, contributing 12 extra repeated rows per model. Within each model, these duplicates are internally consistent: no exact duplicate question has mixed statuses inside the same model. However, Qwen and Llama differ on several duplicate-question outcomes.

Qwen: 12 duplicate questions, 12 extra duplicate rows.

- Out of all options, which segment (Large Cap, Mid Cap etc.) has the most total invested amount?: {'MISMATCH': 2}
- Out of all options, which investment type generates the lowest average dividend income?: {'MISMATCH': 2}
- Among all investors, who has the longest average holding duration across all investors?: {'MISMATCH': 2}
- Group the holdings of investor INV001 by investment type and show the count for each category.?: {'MATCH': 2}
- Calculate the average liquidity rating for investors whose total withdrawal transactions exc...: {'MATCH': 2}
- Do you know which payment source channel handles the highest number of cash flow transactions?: {'MISMATCH': 2}

Llama: 12 duplicate questions, 12 extra duplicate rows.

- Out of all options, which segment (Large Cap, Mid Cap etc.) has the most total invested amount?: {'MISMATCH': 2}
- Out of all options, which investment type generates the lowest average dividend income?: {'MISMATCH': 2}
- Among all investors, who has the longest average holding duration across all investors?: {'MISMATCH': 2}
- Group the holdings of investor INV001 by investment type and show the count for each category.?: {'MISMATCH': 2}
- Calculate the average liquidity rating for investors whose total withdrawal transactions exc...: {'MATCH': 2}
- Do you know which payment source channel handles the highest number of cash flow transactions?: {'MISMATCH': 2}

Actual Correctness Sanity Check

The sanity check reviewed rows where ground truth was machine-checkable as JSON, a number, no rows returned, or an explicit no-data/business-rule statement. It checked whether required values appeared in the answer and treated no-data answers as acceptable only when the ground truth also indicated no data or no rule.

Qwen audit buckets:

- Supported incorrect: 38
- Supported MATCH: 34
- Supported PARTIAL: 10
- MATCH needs manual review: 10

Llama audit buckets:

- Supported incorrect: 68
- Supported MATCH: 12
- Supported PARTIAL: 10
- PARTIAL needs manual review: 2

Caveat: Some MATCH rows answer a no-data ground truth with a numeric zero. This can be correct for count questions, but it is risky for value/list questions unless the evaluation policy explicitly treats missing source data as zero. Those rows should be manually reviewed before publishing a strict benchmark score.

Representative Head-to-Head Examples

Qwen better:

Question: Across the entire investor database, how many Gold holdings does investor INV-005 hold?

Ground truth: There is no data in raw_data_sample to answer this query.

Qwen (MATCH): Investor INV-005 holds 0 Gold holdings.

Llama (PARTIAL): Investor INV-005 holds 0 Gold holdings across the entire investor database.

Question: Across the entire investor database, how many investors affected by the Interest Rate Hike scenario have a Moderate p...

Ground truth: There is no business rule/data to calculate this from raw_data_sample.

Qwen (MATCH): No investors affected by the Interest Rate Hike scenario have a Moderate portfolio health status.

Llama (PARTIAL): There are 0 investors affected by the Interest Rate Hike scenario with a Moderate portfolio health status.

Question: Across the portfolio database, list all redemption events for investor inv-008 with dates and amounts.?

Ground truth: There is no data in raw_data_sample to answer this query.

Qwen (MATCH): No data was found in the knowledge graph. The requested entity may not exist.

Llama (MISMATCH): A database error prevented data retrieval.

Llama better:

Question: Compare the preferred sector in the investor's profile to the sector with the highest allocation percentage in their ...

Ground truth: {"investor_id": "INV-001", "stated_sector_focus": "Government, Gold, Debt", "highest_actual_sector": "Private Equity"...

Qwen (CRASH): ERROR: Operator Compare not found in registry!

Llama (MISMATCH): No data was found in the knowledge graph. The requested entity may not exist.

Question: Summarise the total cash flow amount for each month for investor INV003.?

Ground truth: [{"transaction_month": "2023-04", "total_cash_flow": 5000}]

Qwen (MISMATCH): A database error prevented data retrieval.

Llama (PARTIAL): For investor INV003, the total cash flow amount for April 2023 is

\$5000.

Question: What was the total investment cost made in 2022?

Ground truth: 30560651

Qwen (CRASH): ERROR: Request timed out.

Llama (MISMATCH): The total investment cost made in 2022 was \$0.

Tie MATCH:

Question: Calculate the average liquidity rating for investors whose total withdrawal transactions exceed 50,000.?

Ground truth: There is no business rule/data to calculate this from raw_data_sample.

Qwen (MATCH): No data was found in the knowledge graph. The requested entity may not exist.

Llama (MATCH): No data was found in the knowledge graph. The requested entity may not exist.

Question: Calculate the average liquidity rating for investors whose total withdrawal transactions exceed 50,000.?

Ground truth: There is no business rule/data to calculate this from raw_data_sample.

Qwen (MATCH): No data was found in the knowledge graph. The requested entity may not exist.

Llama (MATCH): No data was found in the knowledge graph. The requested entity may not exist.

Question: Considering all investors in the system, which investor bought the most holdings in 2023?

Ground truth: (no rows returned)

Qwen (MATCH): No data was found in the knowledge graph. The requested entity may not exist.

Llama (MATCH): No data was found in the knowledge graph. The requested entity may not exist.

Tie MISMATCH:

Question: Across the portfolio database, what was the total cash inflow from deposits and sip contributions in the last 12 months?

Ground truth: 0

Qwen (MISMATCH): A database error prevented data retrieval.

Llama (MISMATCH): No data was found in the knowledge graph. The requested entity may not exist.

Question: Among all investors, which ones have not purchased any new holdings in the last 12 months?

Ground truth: [{"investor_id": "INV-001", "investor_name": "Deepak Sharma"}, {"investor_id": "INV-002", "investor_name": "Neha Shar..."}]

Qwen (MISMATCH): A database error prevented data retrieval.

Llama (MISMATCH): A database error prevented data retrieval.

Question: Among all investors, which ones hold Private Equity investments?

Ground truth: [{"investor_id": "INV-001", "investor_name": "Deepak Sharma"}]

Qwen (MISMATCH): No data was found in the knowledge graph. The requested entity may not exist.

Llama (MISMATCH): A database error prevented data retrieval.

Tie PARTIAL:

Question: Given all the data available, which risk profile group has the highest average investment goal target amount?

Ground truth: {"risk_tolerance": "High", "average_target_amount": 1500000}

Qwen (PARTIAL): The risk profile group with the highest average investment goal target amount is "44b3866e-4285-4987-a809-beb00bfc7653" with an average target...

Llama (PARTIAL): The risk profile group with the highest average investment goal target amount is 44b3866e-4285-4987-a809-beb00bfc7653, with a total target amo...

Question: Which portfolio segment has the highest total invested amount across all investors?

Ground truth: {"segment": "Private Equity", "total_invested_amount": 11019100}

Qwen (PARTIAL): The portfolio segment with the highest total invested amount across all investors is Private Equity, with a total amount of 15028100.

Llama (PARTIAL): The portfolio segment with the highest total invested amount across all investors is Private Equity, with a total of \$15,028,100.

Tie CRASH:

Question: Could you show me the total SWP (Systematic Withdrawal Plan) amount per investor.?

Ground truth: [{"investor_id": "INV-003", "total_swp_amount": 5000}]

Qwen (CRASH): ERROR: Request timed out.

Llama (CRASH): ERROR: Request timed out.

Question: I need to find out how many investors have received dividend cash flows sourced from HDFC Bank?

Ground truth: 0

Qwen (CRASH): ERROR: Request timed out.

Llama (CRASH): ERROR: Request timed out.

Question: Identify the sector earned the highest total dividends across all investors?

Ground truth: {"sector": "Technology", "total_dividends": 17084}

Qwen (CRASH): ERROR: Request timed out.

Llama (CRASH): ERROR: Request timed out.

Top Error Patterns

Qwen:

- 4 at Scan: Expected SelectQuery, found 'BIND' (at char 557), (line:17, col:3)
- 2 at Scan: Expected SelectQuery, found 'BIND' (at char 419), (line:12, col:3)
- 2 at Scan: Expected SelectQuery, found 'FILTER' (at char 522), (line:15, col:3)
- 2 at Scan: Cannot eval thing: None (<class 'NoneType'>)
- 2 at Scan: '<' not supported between instances of 'NoneType' and 'NoneType'

Llama:

- 39 at Scan: SPARQL uses terms that do not exist in the RDF graph.
- 17 at Validate: SPARQL uses terms that do not exist in the RDF graph.
- 2 at Scan: 'NotBoundError' object has no attribute 'datatype'
- 2 at Scan: Expected SelectQuery, found 'GROUP' (at char 262), (line:7, col:1)
- 2 at Scan: Expected SelectQuery, found 'GROUP' (at char 463), (line:13, col:3)

Final Conclusion

Qwen is the clear winner in this retest. It is correct much more often, produces many more usable answers when partial credit is included, and has fewer scan-stage failures.

Llama remains mostly incorrect in this run, with a high mismatch rate and many database/SPARQL retrieval failures.

The final hard benchmark should manually review no-data/zero cases and PARTIAL answers. For the current CSV evidence, the most defensible conclusion is: Qwen outperforms Llama, but neither model is fully reliable yet.

Generated from local CSV files only. No external sources were used.